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NIRv and SIF better estimate phenology than NDVI and EVI: Effects of spring and autumn phenology on ecosystem production of planted forests

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ABSTRACT

Phenology plays an important role in affecting carbon sequestration in terrestrial ecosystems in the context of climate change. Remote sensing techniques have been widely used to investigate land surface phenology and the effects of phenology on ecosystem production at regional and global scales. Recently, the near-infrared reflectance of vegetation (NIRv) and solar-induced chlorophyll fluorescence (SIF) have been shown to be more promising metrics of gross primary production (GPP) than the normalized difference vegetation index (NDVI) and the enhanced vegetation index (EVI). However, there is a lack of comparison in the performance of these techniques for deriving phenological metrics. In this study, we explored the consistency in phenological metrics derived from both remote sensing approaches (NDVI, EVI, NIRv, and SIF) and flux tower GPP at six plantations (two broadleaf forests (BF) and four coniferous forests (CF)) in eastern China over the period 2006–2020. The vegetation indices (NDVI, EVI, NIRv) were derived from MODIS data, and SIF was based on the global, OCO-2 based SIF product (GOSIF). We further evaluated the effects of spring and autumn phenology on GPP. The results showed that the flux tower GPP was effectively tracked by NDVI, EVI, NIRv, and SIF ($P < 0.001$). Meanwhile, the phenological metrics derived from EVI, NIRv, and SIF, including the start of growing season (SOS), the end of growing season (EOS), and the length of growing season (GSL), had significant relationships with those derived from GPP at the six plantations ($P < 0.05$). NIRv and SIF were more effective at estimating phenological information than NDVI and EVI. In addition, the root mean squared deviation (RMSD) values between the GPP- and NIRv-retrieved phenological dates were less than those derived from NDVI, EVI, and SIF at the BF sites. However, the differences among RMSD values of NDVI, EVI, NIRv, and SIF were not significant at the CF sites. The linear regression analysis showed that the advance of SOS_{GPP} (i.e., SOS derived from GPP) significantly increased GPP (R^2=0.29, $P < 0.05$) over the period from March to April, and the delay of EOS_{GPP} (i.e., EOS derived from GPP) remarkably enhanced GPP (R^2=0.61, $P < 0.001$) over the period from September to October at the BF sites. The relationship of EOS_{GPP} with GPP (R^2=0.90, $P < 0.05$) over the period September-October was strong at the CF sites. In addition, the variations of annual GPP could be captured by $GSL_{GPP} \times GPP_{max}$, $GSL_{NIRv} \times NIRv_{max}$, and $GSL_{SIF} \times SIF_{max}$ effectively across the BF and CF sites. These findings can help us understand the potential ability of NIRv and SIF in estimating phenological metrics and in revealing the effects of vegetation phenology on the carbon cycle.

1. Introduction

Vegetation phenology is sensitive to climate variability and change and can be used as an indicator of the long-term biological effects of climate change on terrestrial ecosystems (Bradley et al., 1999; Richardson et al., 2013). It is determined by leaf traits and canopy structure such as green-up and leaf flush or canopy function such as photosynthesis (Churkina et al., 2005; Ma et al., 2007; Piao et al., 2007;

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2.5. Phenological metrics retrieval

A sigmoidal logistic function was used to fit the time series of NDVI, EVI, NIRv, SIF, and GPP. The yearly NDVI, EVI, NIRv, SIF, and GPP observations were fitted as:

$$f(t) = a_1 + \frac{a_2}{1 + e^{-\theta_1(t-\beta_1)}} - \frac{a_3}{1 + e^{-\theta_2(t-\beta_2)}} \quad (5)$$

where parameters a_1, a_2, and a_3 represent the minimum value of the seasonal cycle, the amplitudes of the seasonal cycle in the first and second halves, respectively, θ_1 and θ_2 are coefficients determining the speeds of vegetation growth and senescence, β_1 and β_2 are the fitted parameters. SOS was determined with $\beta_1 - 1.317/\theta_1$, and EOS was determined with $\beta_2 + 1.317/\theta_2$ (Zhang et al., 2020) .

2.6. Analysis

Linear regression was used to analyze the relationships of the 8-day averaged NDVI, EVI, NIRv, and SIF with GPP at the six forest sites. To minimize the noise impacts, we used the Savitzky-Golay filter to smooth the NDVI, EVI, NIRv, SIF, and GPP time series. Then, we evaluated linear relationships of NDVI, EVI, NIRv, and SIF with tower GPP in different seasons to compare the ability of the vegetation indices and SIF in capturing the seasonal variability of GPP. In addition, linear regressions were used to explore the effects of spring and autumn phenology on GPP over the period March–April and the period September–October, respectively. The sensitivities of GPP, NDVI, EVI, NIRv, and SIF to the variations of temperature were compared during spring (March–May) and autumn (September–November). The root mean squared deviation (RMSD) was applied to evaluate the differences between the phenological metrics derived from NDVI, EVI, NIRv, and SIF and those derived from tower GPP, and it was calculated as:

$$RMSD = \sqrt{\frac{\sum_{i=1}^{n}(a_i - b_i)^2}{n}} \quad (6)$$

where n is the number of years, a_i represents the phenological metrics obtained from the NDVI, EVI, NIRv, and SIF, and b_i represents the phenological metrics derived from tower GPP data.

3. Results

3.1. Relationships of NDVI, EVI, NIRv, and SIF with GPP

The 8-day averaged NDVI, EVI, NIRv, SIF, and GPP time series in the six planted forests are shown in Fig. 2. There was strong agreement between NDVI, EVI, NIRv, SIF, and GPP. Seasonal dynamics and interannual variation of GPP were effectively captured by NDVI, EVI, NIRv, and SIF. GPP had obvious fluctuations in summer at the JY and MQ sites, and these variations were captured by NDVI, EVI, NIRv, and SIF. The linear relationships of 8-day averaged NDVI, EVI, NIRv, and SIF with GPP at the six forest sites are illustrated in Fig. 3. GPP was strongly correlated with the three vegetation indices and SIF ($P < 0.001$). The relationships were slightly stronger for NIRv (R^2=0.83, $P < 0.001$) than for SIF (R^2=0.80, $P < 0.001$) and EVI (R^2=0.80, $P < 0.001$); the relationship for NDVI (R^2=0.75, $P < 0.001$) was slightly weaker. At the seasonal scale, the linear relationships of NDVI, EVI, NIRv, and SIF with GPP were strong in spring and autumn ($R^2 > 0.70$) (Fig. 4a and c) and were moderate in summer (Fig. 4b).

3.2. Consistency of phenological metrics derived from NDVI, EVI, NIRv, SIF, and GPP

We compared the phenological metrics derived from NDVI, EVI, NIRv, and SIF with those derived from GPP at each site (Fig. 5). There was obvious site-to-site variability of phenology among the satellite products. The SOS derived from EVI, NIRv, and SIF was all later than that derived from GPP, while NDVI had no advance or delay at the JY site. Different discrepancies were found at the SHB site: SOS retrieved from NDVI and NIRv was earlier than that derived from GPP, while SOS retrieved from EVI and SIF was later than that derived from GPP. There were similar discrepancies at MQ, JZ, and HS_1: the satellite-derived SOS was all earlier than that derived from GPP. Opposite patterns were found at HS_2: SOS derived from NDVI, EVI, and NIRv was later than that derived from GPP, while SIF led to earlier SOS than GPP. The largest

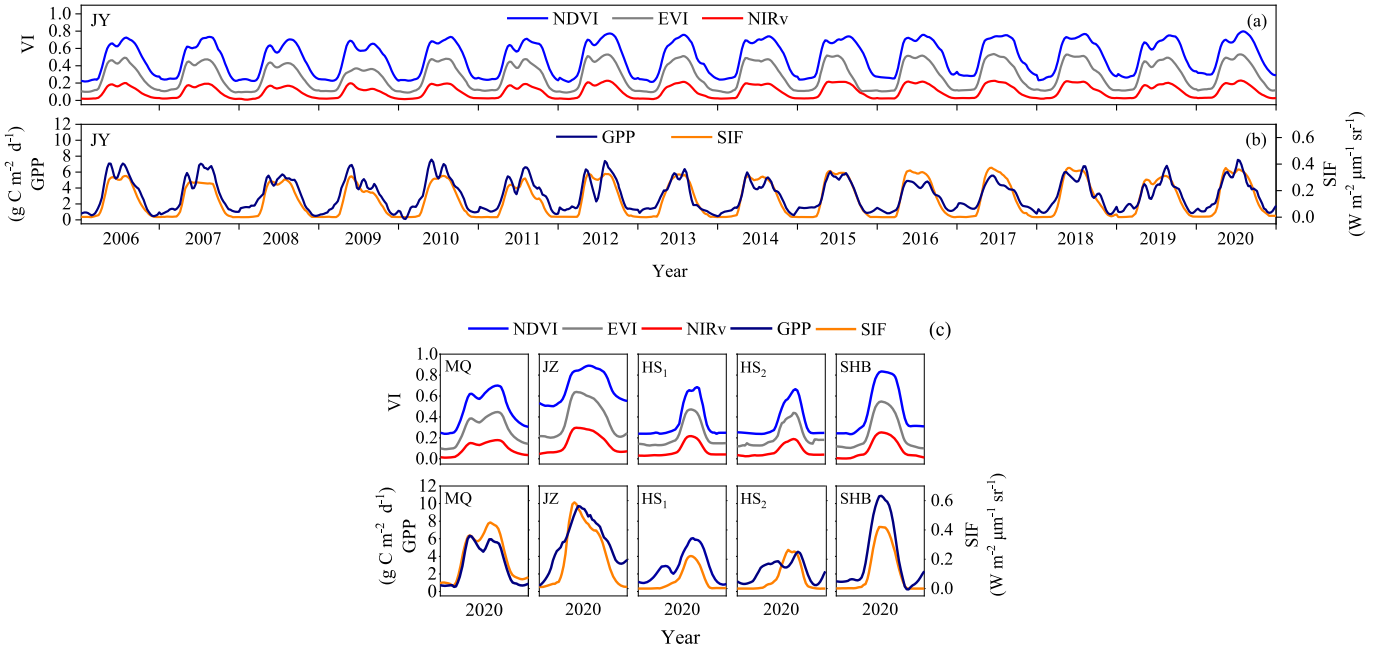


Fig. 2. The seasonal and interannual variations of 8-day averaged normalized difference vegetation index (NDVI), enhanced vegetation index (EVI), and near-infrared reflectance of vegetation (NIRv) at the JY site (a), 8-day averaged solar-induced chlorophyll fluorescence (SIF) and gross primary production (GPP) at the JY site (b) and 8-day averaged NDVI, EVI, NIRv, SIF, and GPP at the MQ, JZ, HS_1, HS_2, and SHB sites (c).

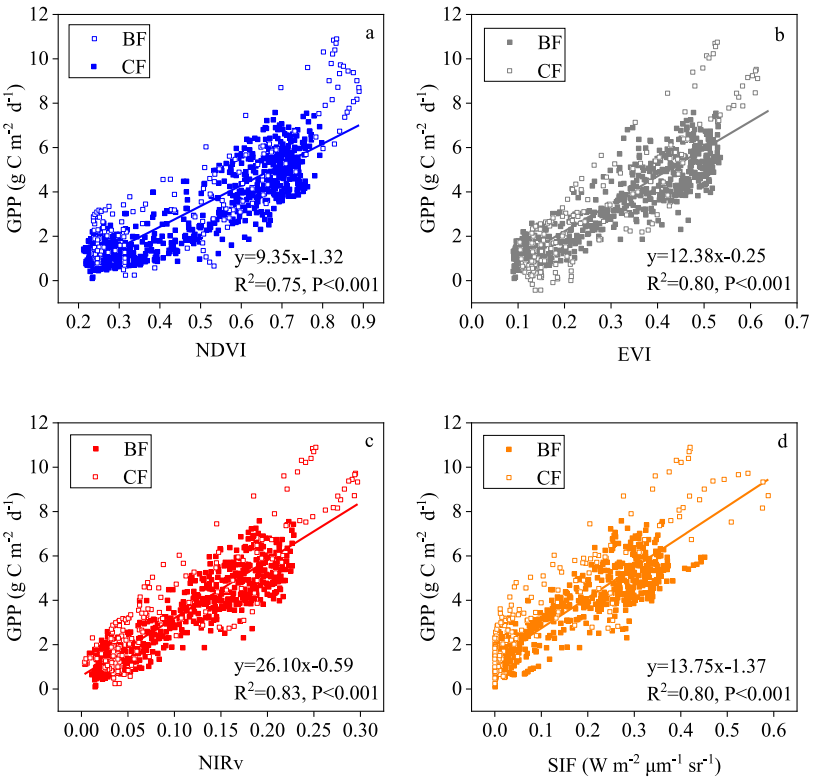


Fig. 3. Relationships of NDVI (a), EVI (b), NIRv (c), and SIF (d) with GPP over six planted forests. BF: the broadleaf forests; CF: the coniferous forests.

discrepancies of SOS were found at the JZ, HS_1, and HS_2 sites. The deviations of EOS found at JY, MQ, and SHB were similar: the EOS derived from the vegetation indices was all later than that derived from GPP, while the EOS derived from SIF was largely earlier than that derived from GPP. On the contrary, the EOS derived from NDVI, EVI, NIRv, and SIF was all earlier than GPP derived EOS at the JZ, HS_1, and HS_2 sites.

The GSL derived from the vegetation indices was longer than that derived from GPP at JY, MQ, SHB, and JZ except that NIRv derived GSL was shorter than GPP derived GSL at the JY site. However, the SIF-derived GSL was shorter than GPP-derived GSL at JY, MQ, and SHB but longer than GPP-derived GSL at the JZ site. At the HS_1 and HS_2 sites, the satellite-derived GSL was shorter than GPP-derived GSL. In a word, the differences of SOS and EOS derived from satellite products at JY, MQ, and SHB were smaller than those at JZ, HS_1, and HS_2. However, the deviations of GSL at each site were not obvious in comparison with those of SOS and EOS.

Phenological metrics derived from NDVI, EVI, NIRv, and SIF were compared with those derived from GPP (Fig. 6). The phenological metrics (i.e., SOS, EOS, and GSL) derived from the three vegetation indices and SIF had significant relationships with those derived from GPP ($P < 0.05$) at the six forest sites with the exception that the relationship between NDVI-derived EOS and GPP-derived EOS was not significant ($P > 0.05$). Both SOS and GSL derived from NDVI, EVI, NIRv, and SIF were strongly correlated with those derived from GPP ($R^2 > 0.72$; $P < 0.001$). NIRv (R^2=0.45; $P < 0.001$) and SIF (R^2=48; $P < 0.001$) performed better in capturing EOS than EVI (R^2=0.26; $P < 0.05$) and NDVI (R^2=0.14; $P > 0.05$) (Fig. 6). This indicates that NIRv and SIF can better estimate phenological metrics than NDVI and EVI.

The RMSD values for the SOS, EOS, and GSL derived from the three vegetation indices and SIF were compared (Fig. 7). At the BF sites, the RMSD values of SOS, EOS, and GSL were 8, 24, and 25 days, respectively, for NDVI; 8, 19, and 18 days, respectively, for EVI; 8, 13, and 12 days, respectively, for NIRv; and 8, 18, and 18 days, respectively, for SIF (Fig. 7a). This indicates that NIRv was slightly more effective in extracting phenological information than SIF, EVI, and NDVI at the BF sites. At the CF sites, the RMSD values of SOS, EOS, and GSL were 23, 19, and 21 days, respectively, for NDVI; 21, 19, and 21 days, respectively, for EVI; 22, 19, and 19 days, respectively, for NIRv; and 25, 22, and 21 days, respectively, for SIF (Fig. 7b). The differences of the RMSD values among NDVI, EVI, NIRv, and SIF were not remarkable at the CF sites.

3.3. Effects of phenology on ecosystem production

Fig. 8 illustrates the effects of SOS_{GPP} and EOS_{GPP} on GPP over the period March-April and the period September-October at the BF and CF sites. Over the period March-April, advancing SOS_{GPP} significantly increased GPP (R^2=0.29; $P < 0.05$). Meanwhile, over the period September-October, the delaying EOS_{GPP} significantly enhanced GPP (R^2=0.61; $P < 0.001$) at the BF sites. At the CF sites, the relationship of SOS_{GPP} with GPP over the period March-April was unremarkable, while GPP (R^2=0.90; $P < 0.05$) over the period September-October was strongly correlated with EOS_{GPP}.

3.4. Joint control of GSL and seasonal peaks on GPP

We compared the ability of the seasonal peaks of NDVI, EVI, NIRv, and SIF and the resulting GSL estimates in explaining the annual GPP at the six plantations (Fig. 9). The two products: $GSL_{NDVI} \times NDVI_{max}$ and $GSL_{EVI} \times EVI_{max}$ were not effective in explaining the variability of annual GPP ($P > 0.05$) (Fig. 9a, b). $GSL_{NIRv} \times NIRv_{max}$ and $GSL_{SIF} \times SIF_{max}$ explained 26% and 38% of the variation of interannual GPP, respectively (Fig. 9c, d). $GSL_{GPP} \times GPP_{max}$ explained 53% of the variation of interannual GPP at the six plantations ($P < 0.001$) (Fig. 9e). GPP_{max} was significantly correlated with annual GPP with R^2=0.80 ($P < 0.001$) (Fig. 9f). However, annual GPP was not significantly correlated with GSL derived from GPP (Fig. 9g), indicating that GPP_{max} had a larger effect on annual GPP than did GSL.

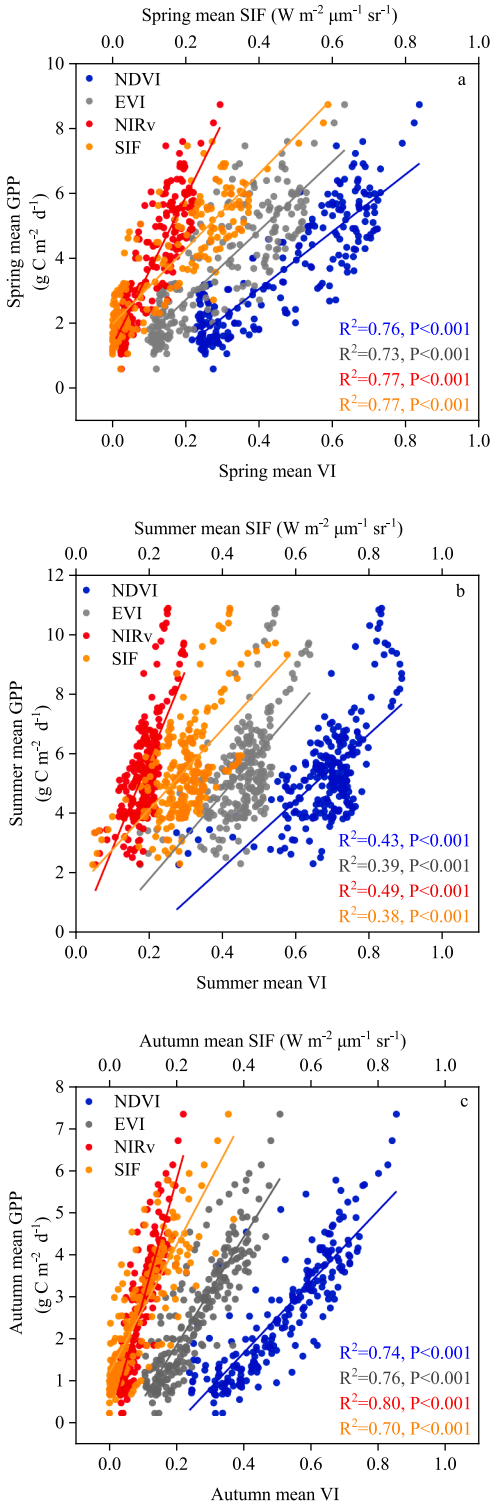


Fig. 4. Relationships of vegetation indices (NDVI, EVI, NIRv) and SIF with GPP in spring (a), summer (b), and autumn (c) over six planted forests.

4. Discussion

4.1. Consistency between NDVI, EVI, NIRv, SIF, and GPP data

The seasonal dynamics and interannual variations of GPP at the six forest plantation sites were effectively captured by the three vegetation indices and SIF, which is consistent with the results of Chang et al. (2019). Our results showed that NIRv, a product of the near-infrared reflectance and the normalized difference vegetation index (Badgley et al., 2017), had a slightly stronger relationship with GPP compared with NDVI and EVI. The results of this study were in agreement with the findings of Huang et al. (2019). They analyzed the linear relationships between vegetation indices and flux tower GPP over 121 flux sites, and pointed out that $NDVI_{BRDF}$, EVI_{BRDF}, and $NIR_{V,\ BRDF}$ explained 56%, 62%, and 70% of the variance in GPP, respectively. Wang et al. (2020b) reported that the R^2 of the linear regressions between GPP and NIRv was higher than that between GPP and NDVI. Wang et al. (2019a) also found strong temporal consistencies between SIF, EVI, NIRv, and GPP over six flux sites. In this study, we showed that at the seasonal scale, the R^2 values of NDVI, EVI, NIRv, and SIF with GPP were lower in summer than in spring and autumn, which is consistent with the result of Ma et al. (2018) that the linear relationship between simulated GPP and satellite-derived SIF in China during 2007–2014 was stronger in spring (R^2=0.60) and autumn (R^2=0.64) than in summer (R^2=0.52).

4.2. Performance of vegetation indices and SIF in extracting phenology

The vegetation indices have been widely used in monitoring phenological information for different vegetation types (Zhang et al., 2003; Kariyeva and Van Leeuwen, 2011; Hou et al., 2014; Verhegghen et al., 2014; Cao et al., 2015; Wang et al., 2017; Wu et al., 2017). Previous studies found that vegetation indices were more effective for extracting phenological stages at deciduous broadleaf forests (Garrity et al., 2011; Melaas et al., 2013; Liu et al., 2016). The three vegetation indices and SIF had identical RMSD values for SOS, while NIRv had slightly smaller RMSD than SIF, EVI, and NDVI for EOS and GSL at the BF sites. SIF, the re-emission of absorbed radiation by chlorophyll molecules (Baker, 2008), can be more effective in capturing seasonal dynamics of photosynthesis (Baker, 2008; Walther et al., 2016; Li et al., 2021). However, there still existed large differences between GPP-based and SIF-based phenology in comparison with NIRv-based phenology, possibly due to the 8-day temporal resolution of SIF that affected the ability of SIF to reflect phenological information (Li and Xiao, 2019).

It is challenging to estimate phenological events of coniferous forests using NDVI and EVI, and the difference between SOS derived from EVI and that derived from GPP was about 68 days in United States (Peng et al., 2017). Liu et al. (2016) also found that neither NDVI nor EVI can robustly estimate SOS or EOS dates for coniferous forests. The inconsistency between canopy greenness and photosynthesis in coniferous forests was mainly because temperature and water availability were two important factors affecting the start of photosynthesis, and it was difficult to observe the shifts of photosynthesis through canopy structural development (Ensminger et al., 2004; Melaas et al., 2013; Chang et al., 2019). The RMSD values of the phenological information derived from vegetation indices at the CF sites were smaller than those of previous studies (Fig. 7b). It may be related to the use of MODIS BRDF-adjusted surface reflectance that can minimize potential seasonal effects of the illumination geometry (Strahler et al., 1999).

The site-to-site variability of phenology derived from satellite products and flux tower GPP was obvious in this study, which is consistent with the findings of Wang et al. (2019a) and D'Odorico et al. (2015). Based on $^{13}CO_2$-pulse-labeling experiments, Keel and Schädel (2010) found that when young leaves were between 10% and 50% expanded, they began to assimilate CO_2, and the rates of leaf expansion and the photosynthetic ability during leaf expansion were different among species. Meanwhile, it is more difficult for satellites to detect changes in greening when evergreen forests have started photosynthesizing (Koike et al., 2001), which may explain the site-to-site differences between satellite and GPP derived phenology (Cernusak et al., 2009; Morecroft et al., 2003). As for the end of the growing season, satellite products mainly rely on capturing the signal of the reducing chlorophyll *a*, which may lead to a decrease of the ratio (chlorophyll *a*/*b*) (Ustin et al., 2009). During this period, the concentration of chlorophyll *b* is still at a stable level and promotes photosynthesis (Koike

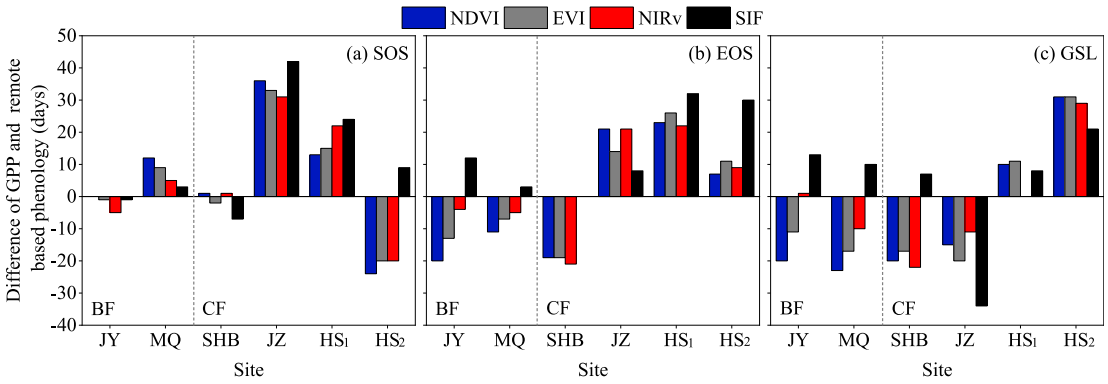


Fig. 5. Differences (in days) of the start of growing season (SOS), the end of growing season (EOS), and the length of growing season (GSL) derived from NDVI, EVI, NIRv, and SIF, relative to that derived from GPP. Negative values indicated that phenology derived from GPP has an earlier SOS, earlier EOS and shorter GSL than that of remote sensing based phenology and vice versa. BF: the broadleaf forests; CF: the coniferous forests. The deviation at the JY site was evaluated using the mean values across years.

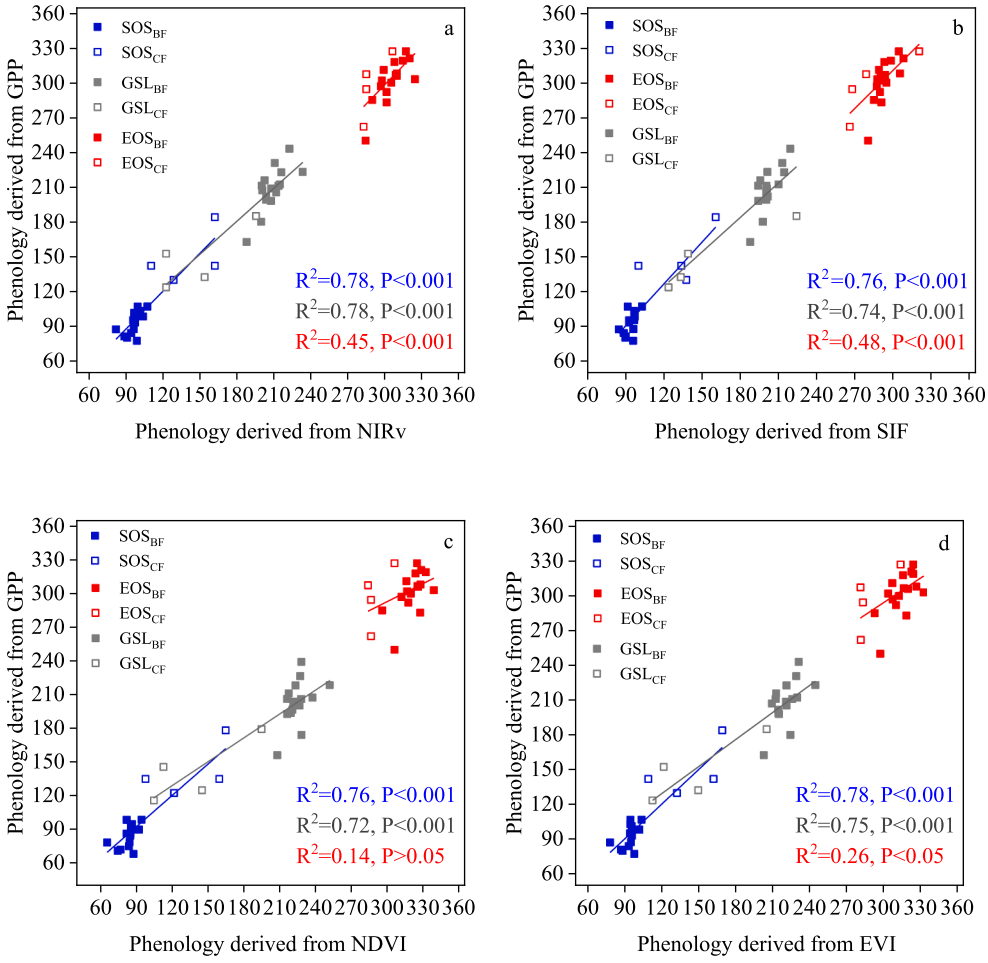


Fig. 6. Relationships of NIRv (a), SIF (b), NDVI (c), and EVI (d) with GPP derived phenological metrics over six planted forests. SOS represents the start of growing season. EOS represents the end of growing season. GSL represents the length of growing season. BF: the broadleaf forests; CF: the coniferous forests.

et al., 2001), and therefore the seasonal differences in chlorophyll *a* content do not always reflect the seasonal differences in photosynthetic capacity among species (Koike, 1990). These reasons may explain the differences between satellite and flux tower GPP derived EOS dates (D'Odorico et al., 2015). Meanwhile, EOS derived from vegetation indices was later than that derived from GPP at the JY and MQ sites (BF sites) likely because chlorophyll degradation was slower than the reduction of photosynthetic activity (Huemmrich et al., 2007). In addition, the timing of EOS derived from SIF was earlier than that derived from GPP at most of the study sites. Similarly,Wang et al. (2020a) reported that EOS derived from SIF was earlier than that derived from GPP for a mixed forest.

Other potential causes may also lead to differences between satellite and GPP derived phenology. In this study, we aggregated the MODIS data to 0.05-degree to match the resolution of the SIF data (i.e., GOSIF). However, the eddy covariance footprints at six planted forest sites of this

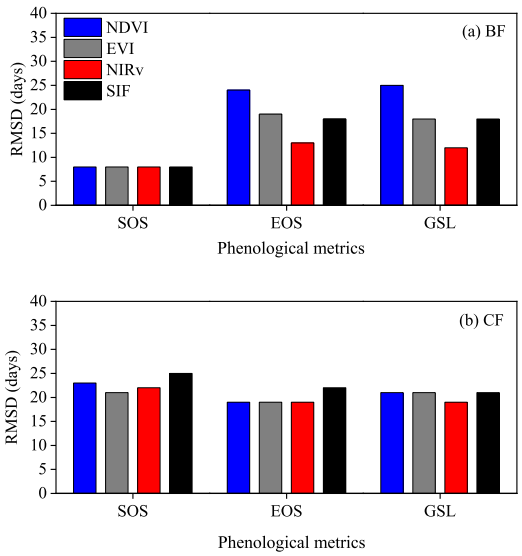


Fig. 7. The root mean squared deviation (RMSD) values of NDVI, EVI, NIRv, and SIF derived phenological metrics in broadleaf forests (BF) (a) and coniferous forests (CF) (b). SOS represents the start of growing season. EOS represents the end of growing season. GSL represents the length of growing season.

study ranged from 2.1 km^2 to 5.1 km^2, and therefore this is a scale mismatch between satellite data and flux tower footprints. Moreover, the vegetation cover is rarely uniform at this scale (Doktor et al., 2009), and the forest stands are often heterogeneous, which makes it difficult to consider the contributions of all the components and to accurately estimate phenological information (Donnelly et al., 2019). Badeck et al. (2004) also pointed out that the mismatches between satellite and ground observations could be due to the heterogeneity in vegetation cover and terrain. Although satellite remote sensing can be an effective tool in estimating phenological information, it is still necessary to evaluate the accuracy of the resulting phenology estimates with ground-based observations (Doktor et al., 2009).

The GPP-derived phenological events (SOS, EOS, and GSL) were strongly correlated with EVI, NIRv, and SIF at the six forest sites, and GPP derived EOS was more strongly related to that derived from SIF and NIRv than to that derived from NDVI and EVI (Fig. 6). This is consistent with the finding of Churkina et al. (2005). Using data from 28 flux sites, Churkina et al. (2005) showed that GSL derived from EVI provided better estimates of the carbon uptake period than that derived from NDVI. Chang et al. (2019) and D'Odorico et al. (2015) also found that for deciduous forests, compared with SOS and EOS derived from NDVI, SOS and EOS estimated from EVI were more strongly correlated with those derived from GPP. Our study shows that NIRv and SIF can better extract phenology information than NDVI and EVI.

Temperature is one of the important factors affecting vegetation phenology (Zhang et al., 2004; Cleland et al., 2006). The rate of chemical reaction inside the vegetation can be promoted with the increase of temperature (Badeck et al., 2004). To understand the differences between NDVI, EVI, NIRv, SIF, and GPP derived phenological metrics, we compared the differences in the relationships of GPP, NDVI, EVI, NIRv, and SIF with temperature in spring and autumn separately (Fig. 10). NDVI, EVI, NIRv, and SIF effectively captured the seasonal variability of vegetation in response to temperature. At the BF sites, the sensitivities of GPP to spring and autumn temperatures were different: the sensitivity of GPP in spring (0.15 g C m^{-2} d^{-1} $°C^{-1}$) was larger than that in autumn (0.14 g C m^{-2} d^{-1} $°C^{-1}$), and the values of GPP in spring started to obviously exceed those in autumn from 11 °C. SIF reflected this variation, but NDVI, EVI, and NIRv were not able to capture this information, and therefore EOS derived from vegetation indices was later than that derived from GPP and SIF at the JY and MQ sites (Fig. 5). At the CF sites, the sensitivities of GPP to spring and autumn temperatures were obviously different. The sensitivity of GPP in autumn (0.28 g C m^{-2} d^{-1} $°C^{-1}$) was larger than that in spring (0.18 g C m^{-2} d^{-1} $°C^{-1}$). Meanwhile, the NDVI, EVI, and NIRv all presented higher sensitivities to temperature in autumn and lower sensitivities in spring, while SIF exhibited a higher sensitivity to temperature in spring and a lower sensitivity in autumn. Similarly, Jeong et al. (2017) pointed out that the temperature sensitivities of the GOSAT and GOME SIF in spring were larger than those in autumn over temperate and boreal forests due to radiation constraint. Therefore, it was necessary to combine the structural and physiological information into GPP models to improve the understanding of the global carbon cycle in the terrestrial ecosystems (Jeong et al., 2017).

4.3. *Effects of phenology on ecosystem production*

Phenology is a sensitive indicator of climate change, and it has a significant impact on ecosystem production (Guo et al., 2020). The warming-induced advance of SOS in spring and delay of EOS in autumn are the two main factors that increase plant production (Tucker et al., 2001; Zhou et al., 2001; Richardson et al., 2010). Richardson et al. (2010) found that phenology was closely correlated with GPP in temperate and boreal forests over 21 FLUXNET sites. Wu et al. (2012) concluded that earlier SOS and later EOS may increase annual GPP, possibly through prolonging GSL at deciduous and evergreen forest sites. Our study showed that SOS and EOS played important roles in determining seasonal variations of GPP (Fig. 8). The advancing SOS_{GPP} significantly increased GPP during the period from March to April at the

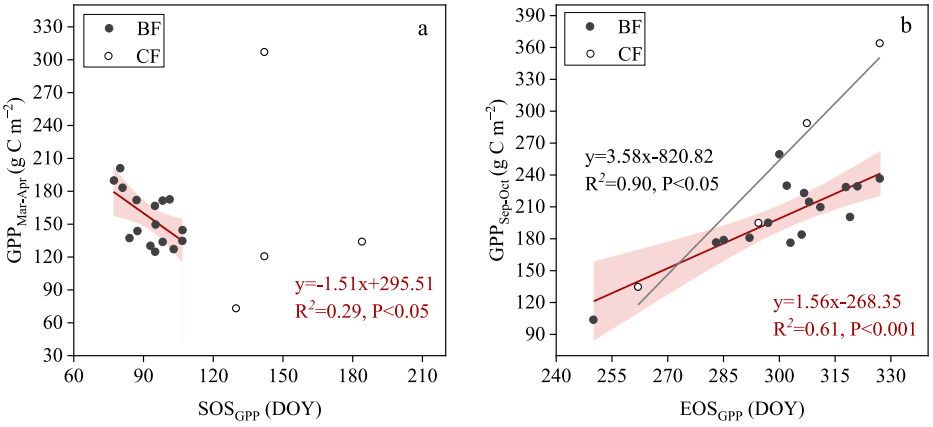


Fig. 8. Relationships of phenological metrics with GPP over the period March–April (a) and September–October (b). SOS represents the start of growing season. EOS represents the end of growing season. BF: the broadleaf forests; CF: the coniferous forests.

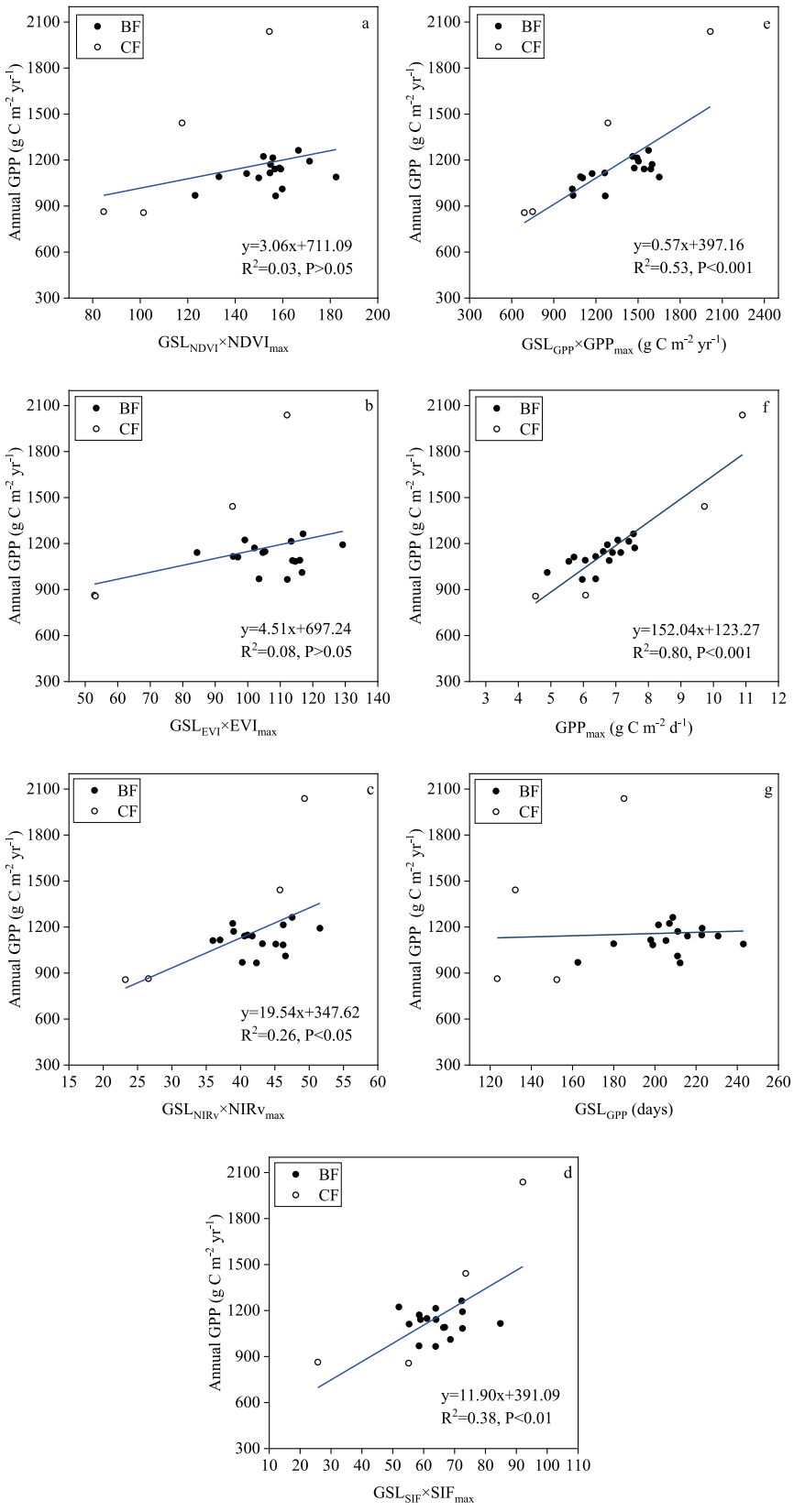


Fig. 9. Relationships of $GSL_{NDVI} \times NDVI_{max}$ (a), $GSL_{EVI} \times EVI_{max}$ (b), $GSL_{NIRv} \times NIRv_{max}$ (c), $GSL_{SIF} \times SIF_{max}$ (d), $GSL_{GPP} \times GPP_{max}$ (e), GPP_{max} (f), and GSL_{GPP} (g) with annual GPP over six planted forests. BF: the broadleaf forests; CF: the coniferous forests.

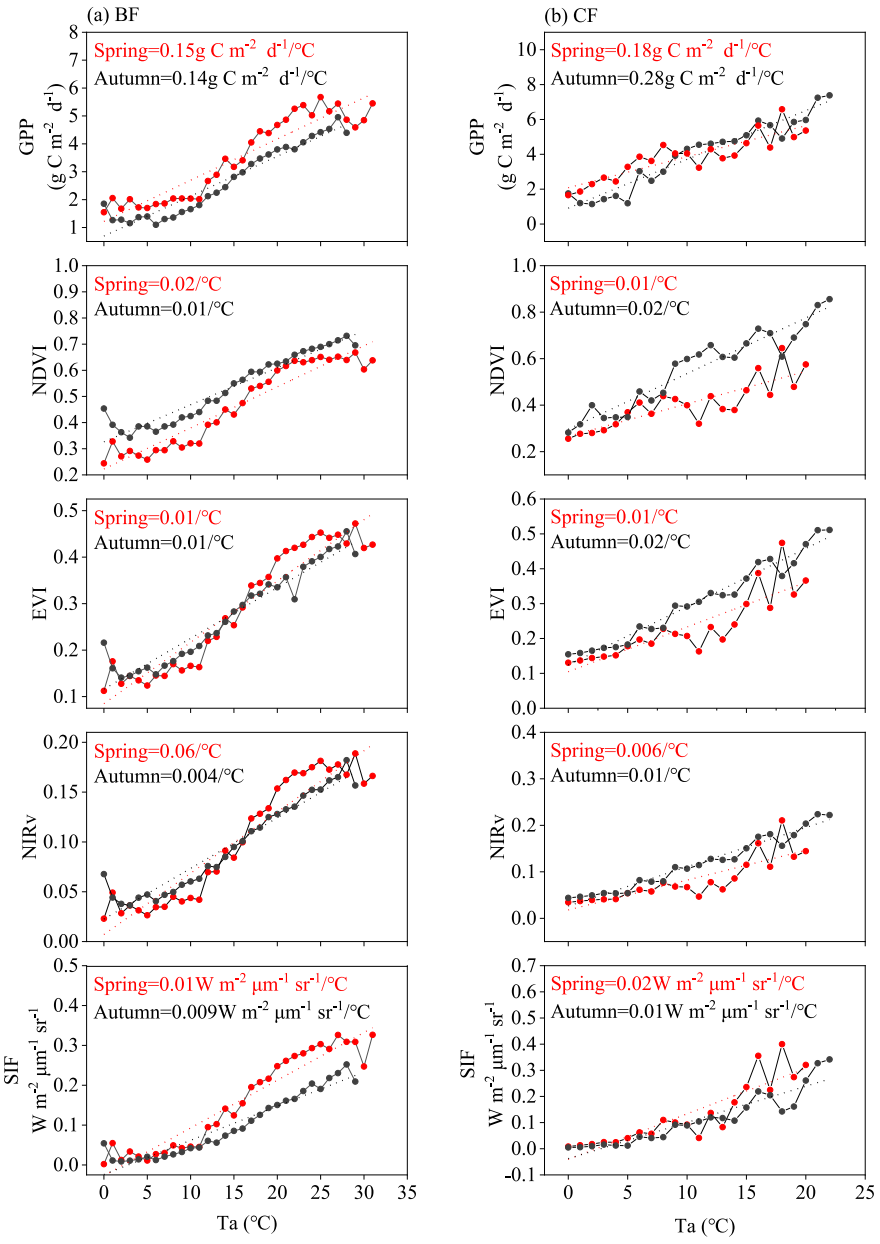


Fig. 10. A comparison of the sensitivity of GPP, NDVI, EVI, NIRv, and SIF to temperature changes during spring (Mar–May) and autumn (Sep–Nov). Each plot illustrates the average value of GPP, NDVI, EVI, NIRv, and SIF for 1 °C temperature interval over the broadleaf forests (BF) (a) and coniferous forests (CF) (b).

BF sites ($P < 0.05$), which is in good agreement with the results illustrated by Du et al. (2019) who found negative correlations between SOS and the anomaly in GPP in grassland and cropland. In addition, our study showed that over the period from September to October, EOS_{GPP} was significantly correlated with GPP at the BF sites. This is in line with the previous study that EOS was positively correlated with net primary productivity in both September and October on the Mongolian Plateau (Bao et al., 2019). However, the relationship of SOS_{GPP} with GPP was not significant over the period March-April, while the relationship of EOS_{GPP} with GPP was remarkable over the period September-October at the CF sites. Richardson et al. (2010) reported that the response of production to phenology in evergreen needleleaf forests was insensitive than that in deciduous broadleaf forests.

4.4. *Explaining temporal GPP variability with the product of GSL and GPP_{max}*

The product of GSL_{GPP} and GPP_{max} effectively captured the variation of annual GPP (R^2=0.53, $P < 0.001$) (Fig. 9). Xia et al. (2015) found that the product of the length of the CO_2 uptake period (CUP) and GPP_{max} explained >90% of the temporal variation of GPP in most areas of North America during the 2000–2010 period. Liu et al. (2021) also pointed out the joint control of growing season length and GPP_{max} explained 87% of the annual GPP in a continental monsoon temperate forest. This suggests that GPP was controlled by both ecosystem-level vegetation phenology and photosynthetic capacity. In addition, we found that GPP_{max} played a more important role in capturing the variation of annual GPP than GSL. The strong relationships between $GSL_{SIF} \times SIF_{max}$ and annual GPP and between $GSL_{NIRv} \times NIRv_{max}$ and annual GPP indicated that NIRv and SIF can be important tools in estimating the interannual variations of GPP.

5. Conclusions

We compared the ability of NDVI, EVI, NIRv, and SIF in capturing the seasonal variability of GPP, investigated the consistency between NDVI, EVI, NIRv, SIF, and GPP derived phenological events, and analyzed the effects of spring phenology and autumn phenology on GPP at the six planted forests from 2006 to 2020. NIRv, EVI, and SIF performed slightly better in capturing the seasonal variability of GPP than NDVI. Meanwhile, NIRv and SIF estimated phenological information more effectively in comparison with NDVI and EVI at the six plantations. The effects of phenology on ecosystem production varied among plant functional types. The advancing SOS_{GPP} significantly increased GPP over the period from March to April, and the delaying EOS_{GPP} remarkably enhanced GPP over the period from September to October at the BF sites. However, the relationship between SOS_{GPP} and GPP over the period March-April was unremarkable at the CF sites. The variation of annual GPP could be captured by $GSL_{GPP} \times GPP_{max}$, $GSL_{NIRv} \times NIRv_{max}$, and $GSL_{SIF} \times SIF_{max}$ effectively across BF and CF sites. Our results highlight the potential of NIRv and SIF in estimating phenological events.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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